ice tests over long periods seem to show that the plates can absorb these strains better than the bolts and this accounts probably for the large number of flexible bolts in use today.

(3) Loose staybolts would not cause corrugation, these bolts are no doubt the results of the plates becoming corrugated and not the cause.

(4) The condition could have been caused by bad water causing scale or solids to accumulate on the water side of the firebox sheet making it necessary for the sheet to be heated to a higher temperature to produce the steam desired, at the same time this increases the difference in temperatures of the side and wrapper sheet, making unequal expansion greater.

Also contributing to the unequal expansion, the firebox side sheet is subjected to cold air every time the firebox door is opened and, on stoker-fired boilers, the air drawn through the grates is not distributed to any particular part of the firebox for there are holes in the fire where cold air enters just below the mud ring and the current of air flows next to the side sheets making it necessary to keep a hot fire or heavy body of fuel for the boiler to generate the steam desired of the power.

(5) I do not believe that redriving the staybolts frequently causes the sheets to corrugate between the rows of bolts.

Inspection of Hollow Staybolts

Q.—Rule 23-B, Interstate Commerce Commission Rules on Inspection of Locomotives reads in part: "On a locomotive with a complete installation of hollow flexible staybolts, caps need not be removed every two years, but the holes must be opened at every hydrostatic test." On some engines, one or two rows of flexible staybolts are behind grates and grate side bars and the holes cannot be opened unless the grates are removed. Is it complying with the law to open all the holes that can be opened with the grates in the firebox and then remove the caps from the bolts behind the grates?—N.T.

A.—Rules 21, 22, and 23 of the laws, rules and instructions for inspection and testing of steam locomotives and tenders and their appurtenances of Interstate Commerce Commission Bureau of Locomotive Inspection, part of which is referred to in the question is as follows:—

(21) *Time of Testing Rigid Bolts.*—All staybolts shall be tested at least once each month. Staybolts shall also be tested immediately after every hydrostatic test.

(22) *Method of Testing Rigid Bolts.*—The inspector must tap each bolt and determine the broken bolts from the sound or the vibration of the sheet. If staybolt tests are made when the boiler is filled with water there must be not less than 50 pounds pressure on the boiler. Should the boiler not be under pressure, the test may be made after draining all water from the boiler, in which case the vibration of the sheet will indicate any unsoundness. The latter test is preferable.

(23) *Method of Testing Flexible Staybolts with Caps.*—Except as provided in paragraph (b), all staybolts having caps over the outer ends shall have the caps removed at least once every two (2) years and the bolts and sleeves examined for breakage. Each time the hydrostatic test is applied the hammer test required by rules 21 and 22 shall be made while the boiler is under hydrostatic pressure not less than the allowed working pressure.

(b) When all flexible staybolts with which any boiler is equipped are provided with a telltale hole not less than $\frac{1}{16}$ inch or more than 7/32 inch in diameter, extending the entire length of the bolt and into the head not less than one-third of its diameter, and these holes are protected from becoming closed by rust and corrosion by copper plating or other approved method, and are opened and tested, each time the hydrostatic test is applied, with an electrical or other instrument approved by the Bureau of Locomotive Inspection, that will positively indicate when the telltale holes are open their entire length, the caps will not be required to be removed. When this test is completed the hydrostatic test must be applied and all staybolts removed which show leakage through the telltale hole. The inner ends of the telltale holes must be kept closed with a fireproof porous material that will exclude foreign matter and permit leakage of steam or water, if the bolt is broken or fractured, into the telltale hole. When this test is completed the ends of the telltale holes shall be closed with material of different color than that removed and a record kept of colors used.

(c) The removal of flexible staybolt caps and other tests shall be reported on the report of inspection form No. 3, and a proper record kept in the office of the railroad company of the inspections and tests made.

(d) Firebox sheets must be carefully examined at least once every month for mudburn, bulging, and indication of broken staybolts.

(e) Staybolt caps shall be removed or any of the above tests made whenever the United States inspector or the railroad company's inspector considers it desirable in order to thoroughly determine the condition of staybolts or staybolt sleeves.

Removing the caps of those staybolts that are behind the grates and grate side bars and testing same as prescribed in the first paragraph of rule 23 would be complying with the law.

In rule 23, paragraph (b) modifies the conditions of the first paragraph to the extent that the flexible staybolt caps do not have to be removed, if the bolts are examined at each hydrostatic test as prescribed in paragraph (b).

Of the two methods, I believe that the removal of the grates and grate side bars and inspecting the staybolts as prescribed in Par. 23 (b) would be a better job and more preferable to removing the staybolt caps.

Reliability of Inside Calking

(*Continued from page 253*)

years, have been inside calking jobs, where riveted construction was used. Furthermore, for new installations inside calking is the rule rather than the exception in specifications.

Referring to Mr. Davies' discussion of the subject, the assertion is made that, in calking, the plate is severely sprung up, but this is not in accordance with good workmanship.

In the first place, the rivets should not be driven unless the butt strap or welt plate is a good fit to the shell. Secondly, the action of driving the rivets will not open up the straps unless excessive pressure is used with very light straps. In fact with good pressures and good fits the joint should come down off the hydraulic riveter with an almost metal to metal contact, and a perfect seal should be obtained by a light calking of the strap edges.

In the third place, the purpose of calking is not to spring the strap edges, but to thicken them up, in order to close the almost imperceptible opening between shell and strap. Furthermore, there should not be any cantilever action. If the joint is tight on the inside strap, the pressure inside the vessel will act uniformly over

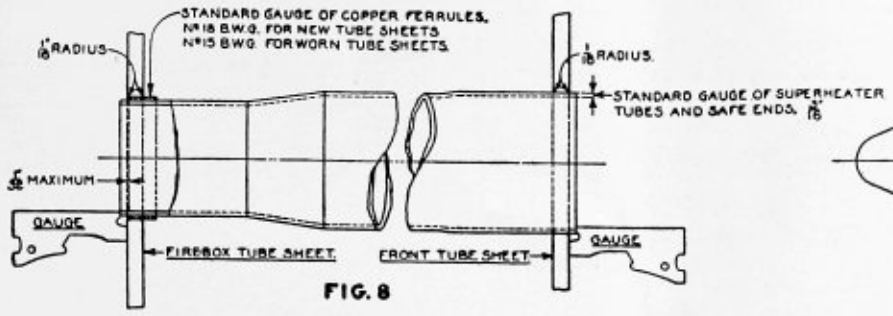


FIG. 8

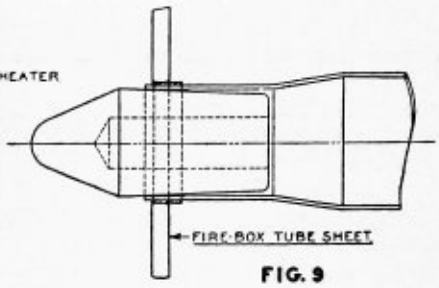


FIG. 9

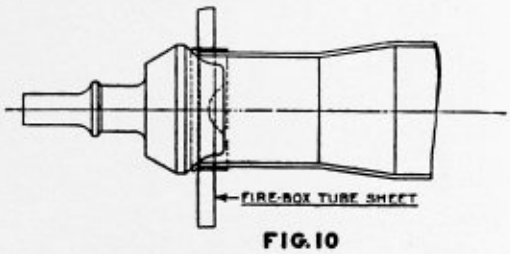


FIG. 10

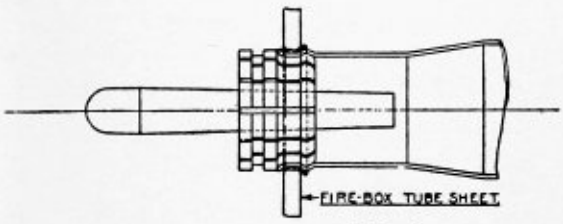


FIG. 11

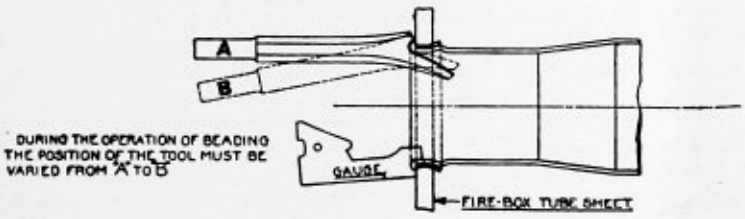


FIG. 12

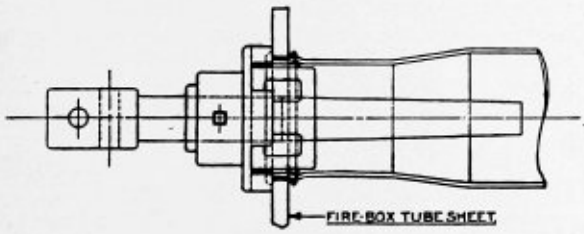


FIG. 13

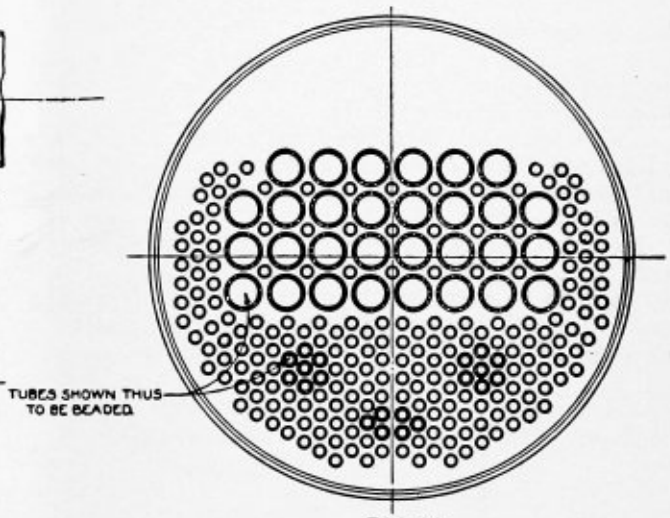


FIG. 14

Method of setting boiler tubes of superheater locomotives

Fig. 8 shows the front and back tube sheets with tube and ferrule in place ready for setting. The tube should project through both sheets ¼ inch as indicated by the gage.

Figs. 9 to 13 inclusive, show the several operations to be performed in setting the tube at the firebox end, and the order in which the work must be performed.

Fig. 9 shows the tube after being pinned, and the pinning tool in place.

Fig. 10 shows the tube after being belled, and the belling tool in place.

Fig. 11 shows the tube after being prossered and the prosser expander in place.

Fig. 12 shows the tube after beading and the beading tool in place. After beading the tube is finished by lightly rolling.

Fig. 13 shows the finished tube and the roller in place. At the front tube sheet the tubes will be pinned, belled, beaded and rolled.

Fig. 14 illustrates a front tube sheet of a superheater locomotive. Small boiler tubes are to be set in the same manner as on saturated steam locomotives, except beaded tubes are to be located as shown in Fig. 14. If there are two parallel vertical center rows there will be but six beaded tubes at the center; on a boiler with a single vertical center row there will be a group of seven.

portion cut out and the good portion used in boilers having shorter tubes.

Tubes that are otherwise good, but have been reduced in weight to less than 1.66 pounds per foot for 2-inch tubes and 1.89 pounds per foot for 2¼-inch tubes, must be scrapped.

Tubes which are slightly pitted or otherwise weakened, but which are still serviceable, shall be kept together in sets so that a complete set of tubes in approximately the same condition may be applied when possible. It is desired that such tubes when re-applied be used in switch power or boilers, the allowed steam pressure of which does not exceed 150 pounds.

Good tubes should be safeended as described below.

Safe ends must be scarfed on one end for half an inch, being tapered to 1/32 inch thickness at end of scarfing, and cut square and smooth on opposite end by the manufacturer.

Tubes to be safe ended must be flared for scarfed end of safe end one half inch in length and opened up an amount equal to the thickness of the tube.

Safe end should then be placed in the end of the tube and both be heated to a white heat, drawn out quickly and welded on a flue welding machine, care being taken to see that the weld is smoothly made.

Small boiler tubes should be safe ended as follows:

First Safe Ending—Apply 5-inch safe end.

Second Safe Ending—Cut off old safe end and apply 8-inch safe end.

Third Safe Ending—Cut off old safe end and apply 12-inch safe end.

Fourth Safe Ending—Apply 5-inch safe end to the

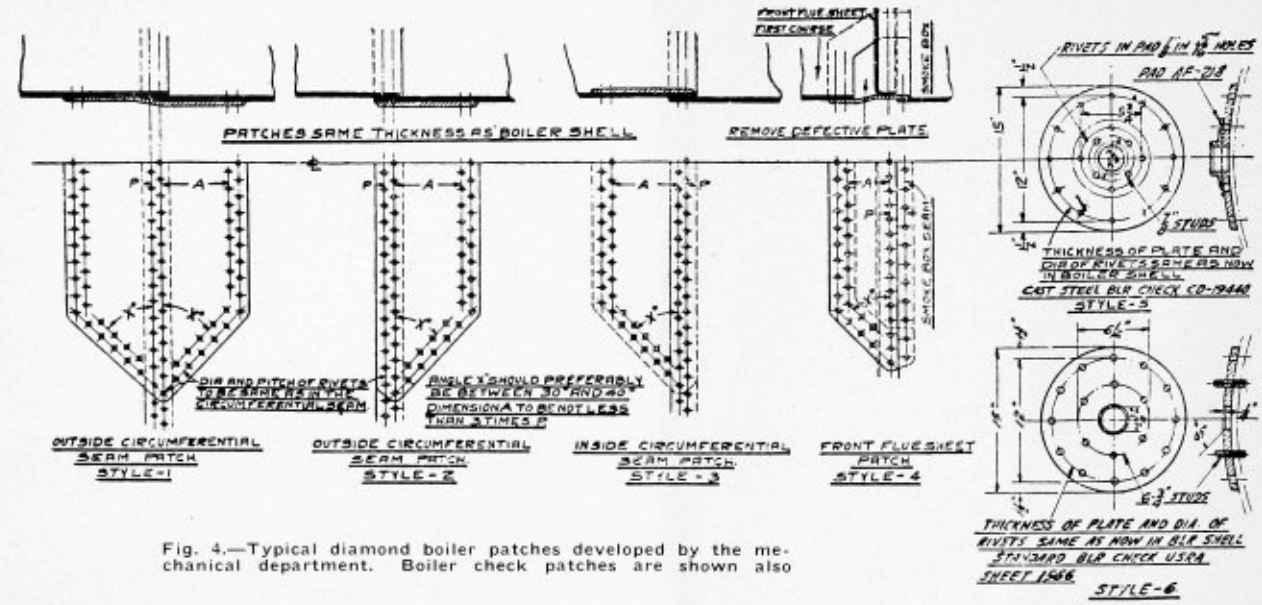


Fig. 4.—Typical diamond boiler patches developed by the mechanical department. Boiler check patches are shown also

diameter for tubes and 1 inch smaller for superheater flues and the scale either filed or ground off the swedge. All flues must be subjected to a hydrostatic test pressure of 250 pounds per square inch or to 90 to 100 pounds air pressure while immersed in water before being applied to the boiler.

In applying, they should be driven through the sheets ¼ inch for superheater flues and $\frac{3}{16}$ inch for all others; rolled tight with standard roller expanders, and at the back flue sheet prosser expanded with standard prosser tool and finally beaded with standard beading tool.

Flues should be prossered by driving in at least twice and turning the expander a distance equal to one-half the width of a section. Bridges of flue sheets should be watched in prossering to avoid overstrain. When flue beads after service are away from the flue sheet, they should be beaded to the sheet and the burr, which forms from constant use of the beading tool, trimmed off.

Flues are to be opened to fit holes in the front flue sheet with a tool provided for that purpose except when flues are applied without the removal of steam pipes at which time flues will be opened after being applied to the boiler; then to be rolled tight with standard roller expanders and superheater flues to be beaded with standard beading tools.

Superheater flues shall be electrically welded to the back flue sheet after having been applied in accordance with the above paragraphs 1 to 4 inclusive.

Small flues should preferably be welded after applying in accordance with the above paragraphs, although local circumstances will govern.

Superheater flues and tubes of engines assigned to the Kent Division, will be expanded and beaded in the usual manner and not electrically welded. When the beads become thin, however, they are to be welded to the back flue sheet.

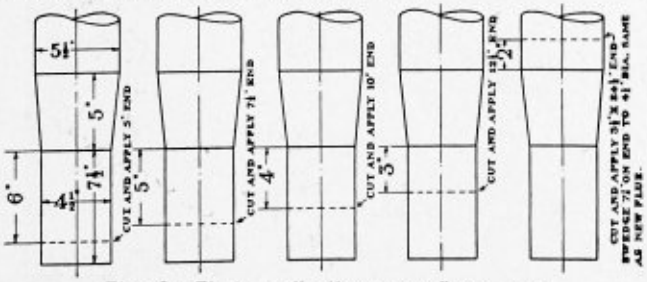


Fig. 5.—First application—new flues; seventh application same as second, etc.

Flue Maintenance: It is required that all flues except those electrically welded to the sheet, be thoroughly prossered at each monthly washout period.

Flue Welding and Preparation: Superheater flues shall contain not more than two welds; and shall be pieced in accordance with Fig. 5. When it becomes necessary to apply a longer end, use the next application as outlined.

Superheater flues shall be swedged to 7½ inches long on the straight and must be annealed after swedging.

Flues, other than superheater, shall contain not more than three welds; methods of piecing being shown by Fig. 6, which will include all flues with 21 inches of welds or less.

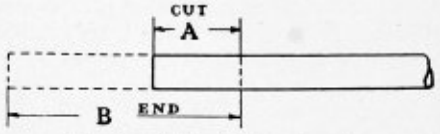


Fig. 6.—Method of piecing flues

	A	B	Remarks
First application	.....	.....	New flue
Second application	1½ inches	5 inches	One weld
Third application	5½ inches	8 inches	One weld
Fourth application	8½ inches	11 inches	One weld
Fifth application	11½ inches	14 inches	One weld
Sixth application	1½ inches	5 inches	Two welds
Seventh application	5½ inches	8 inches	Two welds
Eighth application	8½ inches	11 inches	Two welds
Ninth application	27½ inches	30 inches	One weld
Tenth application	1½ inches	5 inches	Two welds
Eleventh application	5½ inches	8 inches	Two welds
Twelfth application	8½ inches	11 inches	Two welds

Flues (other than superheater), having more than 21 inches of welds shall have all old welds cut off, and a piece of second hand flue welded on of sufficient size to bring it to the required length; and to this shall be added a new 5-inch piece at firebox end. Thereafter the sketch and table to govern.

Flues having worn thin, after being cleaned, having reached the following weight shall not be used further for steam boiler flues:

2 inch diameter, weight 1¾ pounds or less per foot.
2¼ inch diameter, weight 2 pounds or less per foot.
2½ inch diameter, weight 2¼ pounds or less per foot.
3 inch diameter, weight 2¾ pounds or less per foot.
3½ inch diameter, weight 4 pounds or less per foot.

4 inch diameter, weight 4½ pounds or less per foot.
4½ inch diameter, weight 4¾ pounds or less per foot.
5½ inch diameter, weight 7 pounds or less per foot.

Arch Tubes, Application: All scale and burrs must be removed from tube hole and if holes are more than 1/32 inch out of round they must be reamed true. Holes shall not be reamed more than $\frac{3}{16}$ inch above normal size.

Tube is to be cut to correct length, so as to extend through sheet ⅜ inch. All burrs and scale must be removed from ends of tubes by filing or grinding.

In bending, care must be taken to prevent overheating, distortion or kinking of the tube at the bend.

Bends in the tube must be such, that the tube will enter both ends at right angles to the plate and line up with the other arch tubes. Care should be exercised when bending arch tubes, so that the bends will not be of too small a radius to permit the turbo-cleaner to pass through the arch tube.

After the tube has been properly placed, it should be rolled tight with a standard roller expander, belling out end of tube to an angle of 45 degrees.

Arch Tubes—Maintenance: The plates, caps or plugs covering holes in the outer sheet at each end of the arch tube must be removed at each boiler wash.

The interior of the tube must then be thoroughly cleaned of scale, using a water or air turbine cleaner for this purpose. The tube must be thoroughly washed, after cleaning with a turbine cleaner, particular care being taken to see that no scale or sediment lodges in the bottom portion of the tube or at the bend.

Tubes are to be inspected for cracks, bulging, pocketing or distortion due to shifting of bricks each time fire is removed. No tube shall be allowed to remain in service showing any indication of these defects. Calking or plugging of cracks, or of leaks developed at pockets is forbidden.

After tubes have been thoroughly cleaned and inspected, care must be taken in replacing cover plates or caps to see that all joints are in good condition, that studs for plates are proper fit in sheet, nuts properly fitted to studs and a forged plate used. Threads of caps and plugs should be carefully inspected and care exercised to avoid cross threading in their application. A mixture of graphite and oil must be applied to all joints and threads to facilitate subsequent removal.

Staybolts, Water Space Rigid: Water space rigid staybolts shall be made of iron to conform to specification.

When new fireboxes, new half side sheets, new back flue sheets or ¾ door sheets are applied water space rigid bolts shall be $\frac{15}{16}$ inch diameter.

Staybolts should be carried in stock up to and including 1⅛-inch diameter, when larger staybolts than 1⅛-inch diameter are required in roundhouse renewals or $1\frac{1}{16}$-inch in back shop renewals, or new fireboxes applied, holes in outside sheet shall be electric welded and retapped for $\frac{15}{16}$-inch diameter staybolts.

Radial Bolts: Radial bolts shall be made straight, not upset on the ends for threads and of iron to conform to specification. When new fireboxes are applied, radial bolts shall be 1-inch diameter, straight thread.

Crown Bolts: Crown bolts shall be taper head type, with crown sheet fit finished with a taper of 2 inches in 12 inches and of iron to conform to specification. When new fireboxes are applied, crown bolts shall be 1-inch diameter.

Threading All bolts shall be threaded 12 U. S. form threads per inch, and manufactured according to specifications. Care should be exercised in applying bolts to see that threads are not stripped in screwing bolts into sheets and that bolts are a good fit and steam tight.

Staybolts should project through inner and outer sheets $\frac{1}{16}$ inch to allow for formation of head in riveting.

Telltale Holes: Rigid water space bolts shall have telltale holes in outer end ¼ inch diameter and not less than 1⅜ inches deep. Radial bolts shall have telltale holes drilled in outer end 4 inches deep.

Hollow rigid bolts shall be used when stayed area comes back of engine frames, brackets, braces, arch brick or other obstructions.

Staybolt Application: When enlarged holes are reduced by electric weld and retapped, care should be exercised to see that full perfect threads are formed.

Staybolts are not to be applied, when,

(a) Threads in sheet do not show full perfect thread.
(b) Threads on bolts do not show full perfect thread.
(c) Bolt is loose enough to be turned by hand after screwing into second sheet.

Flexible Staybolt Sleeves, Flannery Type: Sleeves for Flannery type flexible staybolts shall be of the unthreaded, electric weld application type.

When threaded type flexible staybolt sleeves are required to be renewed, they shall be replaced by welded type sleeves.

When outside wrapper sheets are applied, holes for regular type welded staybolt sleeves shall be punched 1⅛ inches and for the flush type 2 inches.

The depth of counterbore in the outside wrapper sheet will depend upon the angle of the bore with the sheet. Where bolts are at right angles a width of seat of approximately $\frac{3}{16}$ inch will be found to be ample. Counterboring should be sufficient, however, to give an all around bearing on the curved and angled surfaces of the sleeve. Counterboring should be done with a spindle type combined reamer and counterbore.

When large or complete installations of flexible staybolts are made, the sheets should be held together by applying a flexible staybolt every four or five rows, thus holding the plates parallel while the balance of the bolts are being applied.

The sheets should be thoroughly cleaned of scale and foreign substances, including oil and grease before applying sleeves. The sleeve should then be placed on its bearing and held securely in alinement by an applicator, when being welded. Where sleeves seat at right angles to the firebox sheets, they can be held in place for welding by the head of the flexible bolt. Only sufficient metal to prevent leakage should be applied.

Flexible Staybolt Sleeves, Alco Type: Sleeves for Alco type flexible staybolts shall be of the unthreaded, electric weld application type.

When outside wrapper sheets are applied, holes for regular type welded staybolt sleeves shall be punched 1⅜ inches and for the flush type 2 inches.

The hole in the outside wrapper sheet is to be reamed with a reamer having a taper of 3 inches in 12 inches, and provided with a spindle guide attached.

The hole in the inner sheet should then be tapped, using a bushing guide with proper taper.

The outside wrapper sheet should be thoroughly cleaned of scale, oil, grease or other foreign substance before applying sleeve for welding. After cleaning, the sleeve should be applied, tapping home lightly with a light wooden mallet. The sleeve is then held in place by running in the bolt.

When welding the sleeve, a slip fit cap should be run over the threaded part of the sleeve to protect the threads from sparks during the welding process.

(To be continued)